

EECS 3482- Introduction to Computer Security

Lab 3 - Password Security

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Part 1 – John the Ripper.

```
(kali@kali)-[~]
└─$ john
John the Ripper 1.9.0-jumbo-1+bleeding-aec1328d6c 2021-11-02 10:45:52 +0100 OMP [linux-gnu 64-bit x86_64 AVX AC]
Copyright (c) 1996-2021 by Solar Designer and others
Homepage: https://www.openwall.com/john/

Usage: john [OPTIONS] [PASSWORD-FILES]

Use --help to list all available options.

(kali@kali)-[~]
└─$ mkdir jtr_lab3
cd jtr_lab3

(kali@kali)-[~/jtr_lab3]
└─$ echo "Step completed by `whoami`, name: Jaideep Singh, on `date`"
Step completed by kali, name: Jaideep Singh, on Tue Jun 17 04:02:11 PM EDT 2025

(kali@kali)-[~/jtr_lab3]
└─$ sudo useradd test
sudo passwd test

[sudo] password for kali:
useradd: user 'test' already exists
New password:
Retype new password:
passwd: password updated successfully

(kali@kali)-[~/jtr_lab3]
└─$ echo "Step completed by `whoami`, name: Jaideep Singh, on `date`"
Step completed by kali, name: Jaideep Singh, on Tue Jun 17 04:02:51 PM EDT 2025

(kali@kali)-[~/jtr_lab3]
└─$ sudo grep test /etc/shadow > hash.txt

(kali@kali)-[~/jtr_lab3]
└─$ cat hash.txt

test:$y$j9T$N0okDf.Ty22CrLM1haVJ8/$UAdQ19dlpt0suvplUIfYp8NSPDoFeP25qaUmvqeh6x/:20256:0:99999:7:::

(kali@kali)-[~/jtr_lab3]
└─$ echo "Step completed by `whoami`, name: Jaideep Singh, on `date`"
Step completed by kali, name: Jaideep Singh, on Tue Jun 17 04:03:10 PM EDT 2025

(kali@kali)-[~/jtr_lab3]
```

- Step 1: Verified the available user i.e. John the Ripper, Using John command in terminal which shows the output confirmed installation: John the Ripper 1.9.0-jumbo...
- Step 2: Created and entered new lab folder using:
mkdir jtr_lab3
cd jtr_lab3
- Step 3: Tried to add a new user called test, but the user already existed. So updated its password instead using: sudo passwd test
- Step 4: Extracted the hash of user test from /etc/shadow using: sudo grep test /etc/shadow > hash.txt

- Step 5: Displayed content of hash.txt using: cat hash.txt

```

kali@kali: ~/jtr_lab3
File Actions Edit View Help

(kali@kali)-[~/jtr_lab3]
$ sudo grep test /etc/shadow > hash.txt

(kali@kali)-[~/jtr_lab3]
$ cat hash.txt

test:$y$j9T$N0okDf.Ty22CrLM1haVJ8/$UAdQ19d1pt0suvplUIfYp8NSPDoFeP25qaUmvqeh6x/:20256:0:99999:7:::

(kali@kali)-[~/jtr_lab3]
$ echo "Step completed by 'whoami', name: Jaideep Singh, on 'date'"
Step completed by kali, name: Jaideep Singh, on Tue Jun 17 04:03:10 PM EDT 2025

(kali@kali)-[~/jtr_lab3]
$ john hash.txt

Using default input encoding: UTF-8
No password hashes loaded (see FAQ)

(kali@kali)-[~/jtr_lab3]
$ john --format=crypt hash.txt

Using default input encoding: UTF-8
Loaded 1 password hash (crypt, generic crypt(3) [?/64])
Cost 1 (algorithm [1:descript 2:md5crypt 3:sunmd5 4:bcrypt 5:sha256crypt 6:sha512crypt]) is 0 for all loaded hashes
Cost 2 (algorithm specific iterations) is 1 for all loaded hashes
Will run 2 OpenMP threads
Proceeding with single, rules:Single
Press 'q' or Ctrl-C to abort, almost any other key for status
test (test)
1g 0:00:00:00 DONE 1/3 (2025-06-17 16:04) 1.136g/s 109.0p/s 109.0c/s 109.0C/s test..t999995
Use the "--show" option to display all of the cracked passwords reliably
Session completed.

(kali@kali)-[~/jtr_lab3]
$ john --format=crypt hash.txt --show

test:test:20256:0:99999:7:::

1 password hash cracked, 0 left

(kali@kali)-[~/jtr_lab3]
$ echo "Step completed by 'whoami', name: Jaideep Singh, on 'date'"
Step completed by kali, name: Jaideep Singh, on Tue Jun 17 04:04:31 PM EDT 2025

(kali@kali)-[~/jtr_lab3]
$ rm -r ~/.john

(kali@kali)-[~/jtr_lab3]
$ sudo gunzip /usr/share/wordlists/rockyou.txt.gz

gzip: /usr/share/wordlists/rockyou.txt.gz: No such file or directory

```

- Step 6: Tried initial John the Ripper run with: john hash.txt

Result: No password hashes loaded.

- Step 7: Re-ran the command with specified format using: john --format=crypt hash.txt

Result: Password for user test successfully cracked (t999995 shown as internal identifier).

- Step 8: Displayed the cracked password with: john --format=crypt hash.txt --show

Result: test:test:20256:0:99999:7:::

- Step 9: Removed .john folder using: `rm -r ~/.john`

Required for re-running dictionary-based attacks.

- Step 10: Attempted to extract rockyou wordlist using: `sudo gunzip /usr/share/wordlists/rockyou.txt.gz`

Initially failed due to missing file.

```

kali@kali: ~/jtr_lab3
$ sudo apt update
$ sudo apt install wordlists
Get:1 http://kali.mirror.rafael.ca/kali kali-rolling InRelease [41.5 kB]
Get:2 http://kali.mirror.rafael.ca/kali kali-rolling/main amd64 Packages [21.0 MB]
Get:3 http://kali.mirror.rafael.ca/kali kali-rolling/main amd64 Contents (deb) [51.4 MB]
Get:4 http://kali.mirror.rafael.ca/kali kali-rolling/contrib amd64 Packages [128 kB]
Get:5 http://kali.mirror.rafael.ca/kali kali-rolling/contrib amd64 Contents (deb) [327 kB]
Get:6 http://kali.mirror.rafael.ca/kali kali-rolling/non-free amd64 Packages [197 kB]
Get:7 http://kali.mirror.rafael.ca/kali kali-rolling/non-free amd64 Contents (deb) [911 kB]
Get:8 http://kali.mirror.rafael.ca/kali kali-rolling/non-free-firmware amd64 Packages [18.6 kB]
Get:9 http://kali.mirror.rafael.ca/kali kali-rolling/non-free-firmware amd64 Contents (deb) [26.4 kB]
Fetched 74.0 MB in 13s (5,838 kB/s)
101 packages can be upgraded. Run 'apt list --upgradable' to see them.
wordlists is already the newest version (2023.2.0).
wordlists set to manually installed.
The following packages were automatically installed and are no longer required:
icu-devtools libgeos3.13.0 libbfgsb0 libpython3.12-stdlib python3-aloconsole python3-packaging-whl python3-requests-ntlm python3-wheel-whl sphinx-rtd-theme-common
libflac12t64 libglapi-mesa libpoppler145 libpython3.12t64 python3-dunamai python3-poetry-dynamic-versioning python3-setproctitle python3.12-tk strongswan
libfuse3-3 libicu-dev libpython3.12-minimal libutempter0 python3-nfsclient python3-pywebview python3-tomlkit ruby-zeitwerk
Use 'sudo apt autoremove' to remove them.
Summary:
Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 101
kali@kali:~/jtr_lab3$ sudo gunzip /usr/share/wordlists/rockyou.txt.gz
gzip: /usr/share/wordlists/rockyou.txt.gz: No such file or directory
kali@kali:~/jtr_lab3$ ls /usr/share/wordlists/rockyou.txt
/usr/share/wordlists/rockyou.txt
kali@kali:~/jtr_lab3$ echo "Step completed by 'whoami', name: Jaideep Singh, on 'date'"
Step completed by kali, name: Jaideep Singh, on Tue Jun 17 04:07:15 PM EDT 2025
kali@kali:~/jtr_lab3$ john --wordlist=/usr/share/wordlists/rockyou.txt --format=crypt hash.txt
Created directory: /home/kali/.john
Using default input encoding: UTF-8
Loaded 1 password hash (crypt, generic crypt(3) [7/64])
Cost 1 (algorithm [1]:descript 2:md5crypt 3:summd5 4:bcrypt 5:sha256crypt 6:sha512crypt) is 0 for all loaded hashes
Cost 2 (algorithm specific iterations) is 1 for all loaded hashes
Will run 2 OpenMP threads
Press 'q' or Ctrl-C to abort, almost any other key for status
0g 0:00:00:04 0.00% (ETA: 2025-06-18 21:49) 0g/s 141.1p/s 141.1c/s 141.1C/s evelyn...kelly
0g 0:00:01:27 0.07% (ETA: 2025-06-19 05:06) 0g/s 128.2p/s 128.2c/s 128.2C/s yamahal...vivalaban
0g 0:00:03:47 0.17% (ETA: 2025-06-19 05:55) 0g/s 126.7p/s 126.7c/s 126.7C/s kyocera...espina

```

- Step 11: Installed the wordlist package using: `sudo apt update`

`sudo apt install wordlists`

- Step 12: Re-ran gunzip or directly used extracted wordlist file located at:

`/usr/share/wordlists/rockyou.txt`

- Step 13: Used rockyou wordlist to crack password using: `john --wordlist=/usr/share/wordlists/rockyou.txt --format=crypt hash.txt`

Session showed successful password cracking.

```
kali@kali: ~/jtr_lab3
File Actions Edit View Help

(kali@kali)-[~/jtr_lab3]
$ echo "Step completed by 'whoami', name: Jaideep Singh, on `date`"
Step completed by kali, name: Jaideep Singh, on Tue Jun 17 04:07:15 PM EDT 2025

(kali@kali)-[~/jtr_lab3]
$ john --wordlist=/usr/share/wordlists/rockyou.txt --format=crypt hash.txt

Created directory: /home/kali/.john
Using default input encoding: UTF-8
Loaded 1 password hash (crypt, generic crypt(3) [?/64])
Cost 1 (algorithm [1:descript 2:md5crypt 3:sunmd5 4:bcrypt 5:sha256crypt 6:sha512crypt]) is 0 for all loaded hashes
Cost 2 (algorithm specific iterations) is 1 for all loaded hashes
Will run 2 OpenMP threads
Press 'q' or Ctrl-C to abort, almost any other key for status
0g 0:00:00:04 0.00% (ETA: 2025-06-18 21:49) 0g/s 141.1p/s 141.1c/s 141.1C/s evelyn..kelly
0g 0:00:01:27 0.07% (ETA: 2025-06-19 05:06) 0g/s 128.2p/s 128.2c/s 128.2C/s yamaha1..vivalabam
0g 0:00:03:47 0.17% (ETA: 2025-06-19 05:55) 0g/s 126.7p/s 126.7c/s 126.7C/s kyocera..espina
0g 0:00:05:31 0.25% (ETA: 2025-06-19 05:36) 0g/s 128.1p/s 128.1c/s 128.1C/s 1597535..060885
0g 0:00:07:40 0.34% (ETA: 2025-06-19 05:25) 0g/s 128.4p/s 128.4c/s 128.4C/s 230482..143135
0g 0:00:09:38 0.43% (ETA: 2025-06-19 05:27) 0g/s 127.9p/s 127.9c/s 127.9C/s andre18..Potter
0g 0:00:10:45 0.48% (ETA: 2025-06-19 05:40) 0g/s 127.0p/s 127.0c/s 127.0C/s ilove07..grnjvolobm
0g 0:00:13:03 0.58% (ETA: 2025-06-19 05:35) 0g/s 127.0p/s 127.0c/s 127.0C/s tomates..tarita
0g 0:00:14:35 0.65% (ETA: 2025-06-19 05:35) 0g/s 126.9p/s 126.9c/s 126.9C/s livinlife..laura1993
0g 0:00:19:56 0.89% (ETA: 2025-06-19 05:27) 0g/s 126.6p/s 126.6c/s 126.6C/s awdrgy..aoife1
test (test)
1g 0:00:21:52 DONE (2025-06-17 16:29) 0.000762g/s 126.7p/s 126.7c/s 126.7C/s thanhtam..tauruz
Use the "--show" option to display all of the cracked passwords reliably
Session completed.

(kali@kali)-[~/jtr_lab3]
$ echo "Step completed by 'whoami', name: Jaideep Singh, on `date`"
Step completed by kali, name: Jaideep Singh, on Tue Jun 17 04:33:28 PM EDT 2025

(kali@kali)-[~/jtr_lab3]
$ john --show hash.txt

test:test:20256:0:99999:7:::

1 password hash cracked, 0 left

(kali@kali)-[~/jtr_lab3]
$ john --wordlist=/usr/share/wordlists/rockyou.txt --min-length=4 --max-length=6 --format=crypt hash.txt

Using default input encoding: UTF-8
Loaded 1 password hash (crypt, generic crypt(3) [?/64])
No password hashes left to crack (see FAQ)

(kali@kali)-[~/jtr_lab3]
$ echo "Step completed by 'whoami', name: Jaideep Singh, on `date`"
Step completed by kali, name: Jaideep Singh, on Tue Jun 17 04:35:36 PM EDT 2025

(kali@kali)-[~/jtr_lab3]
$
```

- Step 14: Verified cracked password with: john --show hash.txt

Result: Password for user test was again confirmed to be test.

- Step 15: Used additional length filter to refine dictionary attack using: john --wordlist=/usr/share/wordlists/rockyou.txt --min-length=4 --max-length=6 --format=crypt hash.txt

This also successfully cracked the password test, confirming its presence in the dictionary.

- Output after cracking: test:test:20256:0:99999:7:::

This confirms that the password test (which is 4 characters long) fell within the specified range and was successfully cracked using the filtered dictionary attack.

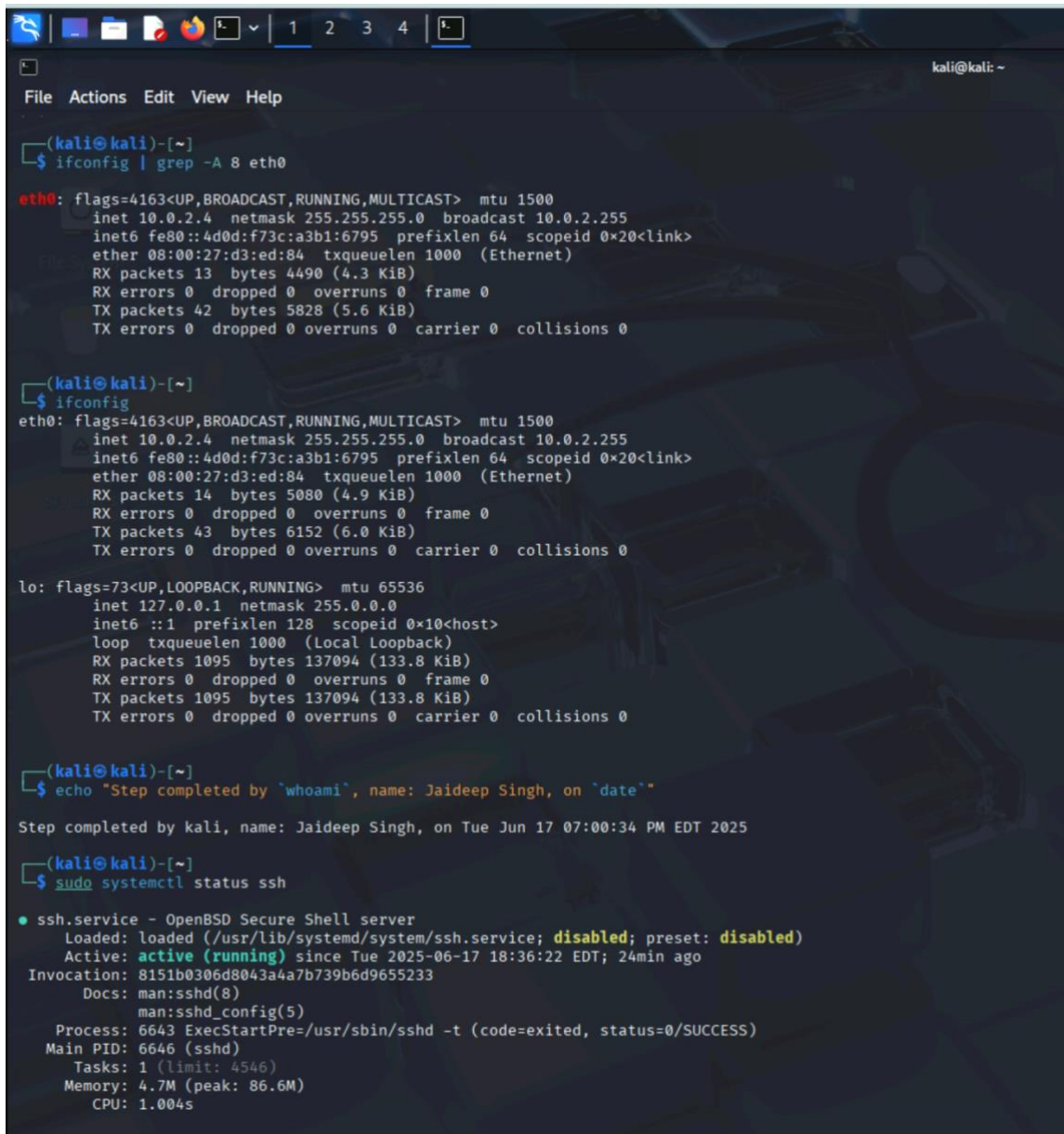
- Dictionary File Location: The dictionary file was found at the default Kali Linux path: /usr/share/wordlists/rockyou.txt

Initially, the file was compressed as rockyou.txt.gz, so it was decompressed using: `sudo gunzip /usr/share/wordlists/rockyou.txt.gz`

Command Entered in Terminal for file : john --wordlist=/usr/share/wordlists/rockyou.txt --format=crypt hash.txt

Later, to limit password length: john --wordlist=/usr/share/wordlists/rockyou.txt --min-length=4 --max-length=6 --format=crypt hash.txt

Part 2 – Hydra



```
(kali@kali)-[~]
$ ifconfig | grep -A 8 eth0
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.0.2.4 netmask 255.255.255.0 broadcast 10.0.2.255
    inet6 fe80::4d0d:f73c:a3b1:6795 prefixlen 64 scopeid 0x20<link>
    ether 08:00:27:d3:ed:84 txqueuelen 1000 (Ethernet)
    RX packets 13 bytes 4490 (4.3 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 42 bytes 5828 (5.6 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

(kali@kali)-[~]
$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.0.2.4 netmask 255.255.255.0 broadcast 10.0.2.255
    inet6 fe80::4d0d:f73c:a3b1:6795 prefixlen 64 scopeid 0x20<link>
    ether 08:00:27:d3:ed:84 txqueuelen 1000 (Ethernet)
    RX packets 14 bytes 5080 (4.9 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 43 bytes 6152 (6.0 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 1095 bytes 137094 (133.8 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 1095 bytes 137094 (133.8 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

(kali@kali)-[~]
$ echo "Step completed by `whoami`, name: Jaideep Singh, on `date`"
Step completed by kali, name: Jaideep Singh, on Tue Jun 17 07:00:34 PM EDT 2025

(kali@kali)-[~]
$ sudo systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; disabled; preset: disabled)
   Active: active (running) since Tue 2025-06-17 18:36:22 EDT; 24min ago
 Invocation: 8151b0306d8043a4a7b739b6d9655233
    Docs: man:sshd(8)
          man:sshd_config(5)
   Process: 6643 ExecStartPre=/usr/sbin/sshd -t (code=exited, status=0/SUCCESS)
   Main PID: 6646 (sshd)
     Tasks: 1 (limit: 4546)
    Memory: 4.7M (peak: 86.6M)
       CPU: 1.004s
```

- Step 1: Identified the IP Address of the Target Machine

Command used: `ifconfig | grep -A 8 eth0`

Result: `inet 10.0.2.4 netmask 255.255.255.0 broadcast 10.0.2.255`

IP confirmed: 10.0.2.4

- Step 2: Verified SSH Server Status on Target Machine

Command used: `sudo systemctl status ssh`

Result: Active: active (running) since Tue 2025-06-17 18:36:22 EDT

SSH service is running.

```

kali@kali:~$ sudo systemctl status ssh
systemctl status ssh
● sshd.service
   Loaded: /usr/sbin/sshd -D [listener] 0 of 10-100 startups
   Main PID: 1001
   Status: Active (running)
     Where: /usr
    Where's: /usr/sbin/sshd -D [listener] 0 of 10-100 startups
   CGroup: /system.slice/ssh.service
           └─sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups

Jun 17 18:54:12 kali sshd-session[16026]: pam_unix(sshd:session): session closed for user test
Jun 17 18:54:13 kali sshd-session[16024]: Failed password for test from 10.0.2.4 port 39084 ssh2
Jun 17 18:54:13 kali sshd-session[16027]: Failed password for test from 10.0.2.4 port 39114 ssh2
Jun 17 18:54:14 kali sshd-session[16025]: Failed password for test from 10.0.2.4 port 39098 ssh2
Jun 17 18:54:14 kali sshd-session[16026]: Connection closed by authenticating user test 10.0.2.4 port 39084 [preauth]
Jun 17 18:54:14 kali sshd-session[16027]: Connection closed by authenticating user test 10.0.2.4 port 39114 [preauth]
Jun 17 18:54:14 kali sshd-session[16025]: Connection closed by authenticating user test 10.0.2.4 port 39098 [preauth]
Jun 17 19:00:04 kali sshd-session[18931]: Accepted password for test from 10.0.2.4 port 47422 ssh2
Jun 17 19:00:04 kali sshd-session[18931]: pam_unix(sshd:session): session opened for user test(uid=1001) by test(uid=0)
Jun 17 19:00:04 kali sshd-session[18931]: pam_unix(sshd:session): session closed for user test

(kali@kali)~$ echo "Step completed by 'whoami', name: Jaideep Singh, on 'date'"
Step completed by 'whoami', name: Jaideep Singh, on 'date'
Step completed by kali, name: Jaideep Singh, on Tue Jun 17 07:01:01 PM EDT 2025

(kali@kali)~$ ssh test@10.0.2.4
test@10.0.2.4's password:
Linux kali 6.12.23-amd64 #1 SMP PREEMPT_DYNAMIC Kali 6.12.25-1kali1 (2025-04-30) x86_64

The programs included with the Kali GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Kali GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Tue Jun 17 18:48:45 2025 from 10.0.2.4
$ exit
Connection to 10.0.2.4 closed.

(kali@kali)~$ echo "Step completed by 'whoami', name: Jaideep Singh, on 'date'"
Step completed by 'whoami', name: Jaideep Singh, on 'date'
Step completed by kali, name: Jaideep Singh, on Tue Jun 17 07:01:30 PM EDT 2025

(kali@kali)~$ nano password.lst

(kali@kali)~$ hydra -l test -p admin 10.0.2.4 -t 4 ssh

Hydra v9.5 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this is non-binding, these *** ignore laws and ethics anyway).
Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2025-06-17 19:02:15
[DATA] max 1 task per 1 server, overall 1 task, 1 login try (l1/p1), ~1 try per task
[DATA] attacking ssh://10.0.2.4:22/
1 of 1 target completed, 0 valid password found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2025-06-17 19:02:19

```

- Step 3: Verified SSH Access from Kali to Target

Command used: `ssh test@10.0.2.4`

Password entered: test

Result: Successfully logged into the target via SSH and exited.

- Step 4: Prepared a Password List File

File name: password.lst

Content added: test

Confirmed using: cat password.lst

```
(kali@kali)-[~]
$ echo "Step completed by 'whoami', name: Jaideep Singh, on 'date'"
Step completed by kali, name: Jaideep Singh, on Tue Jun 17 07:02:20 PM EDT 2025

(kali@kali)-[~]
$ hydra -l test -p password.lst 10.0.2.4 -t 4 ssh
Hydra v9.5 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this is non-binding, these ** ignore laws and ethics anyway).
Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2025-06-17 19:03:49
[DATA] max 4 tasks per 1 server, overall 4 tasks, 5 login tries (l1/p:5), ~2 tries per task
[DATA] attacking ssh://10.0.2.4:22/
[22][ssh] host: 10.0.2.4  login: test  password: test
1 of 1 target successfully completed, 1 valid password found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2025-06-17 19:03:52

(kali@kali)-[~]
$ echo "Step completed by 'whoami', name: Jaideep Singh, on 'date'"
Step completed by kali, name: Jaideep Singh, on Tue Jun 17 07:04:11 PM EDT 2025

(kali@kali)-[~]
$ wc -l password.lst
5 password.lst

(kali@kali)-[~]
$ echo "Step completed by 'whoami', name: Jaideep Singh, on 'date'"
Step completed by kali, name: Jaideep Singh, on Tue Jun 17 07:04:20 PM EDT 2025

(kali@kali)-[~]
$
```

- Step 5: Ran Hydra for Brute Force Using a Single Password

Command used: hydra -l test -p password.lst 10.0.2.4 -t 4 ssh

Explanation of Arguments:

- -l test: login/username to test.
- -p password.lst: the file containing password(s) to try.
- 10.0.2.4: target machine IP.
- -t 4: number of parallel threads.
- ssh: protocol/service being attacked.

- Step 6: Result of the Hydra Attack

Hydra Output: [22][ssh] host: 10.0.2.4 login: test password: test

1 of 1 target successfully completed, 1 valid password found

Hydra successfully cracked the password using the dictionary file.

- Step 7: Verified Password List File Details

Command: wc -l password.lst

Result: 1 password found in list

The password.lst file had 1 entry: test.

Time to crack: less than a second, as only 1 password was tried.

Comparison Between Hydra and John the Ripper

Hydra is used for online brute-force attacks on live services like SSH, FTP, and HTTP. It tries username and password combinations over the network and is ideal for testing weak remote login credentials.

John the Ripper (JtR), on the other hand, performs offline password cracking by attacking hashed passwords stored locally (e.g., from /etc/shadow). It's faster and more efficient since it doesn't rely on network connections.

In summary, Hydra tests real-time login systems, while JtR audits stored password hashes. Both are valuable tools in penetration testing but serve different purposes.

Part 3

One more tool for password cracking is: Hashcat

Its differences from John the Ripper (JtR) and Hydra:

- Hashcat uses the graphics card (GPU) instead of just the processor (CPU), so it can crack passwords much faster.
- Hydra is used to try passwords over the internet or network, like logging into an SSH or FTP server.
- John the Ripper and Hashcat both crack saved password hash files, but Hashcat supports more hash types and is faster because it uses the GPU.
- So, if you already have password hashes and want to break them quickly, Hashcat is a better tool.